Nest Density distribution

2026-06-29

1. Fit nest distribution curve

Nest distance-from-the-coast follows a certain distribution where immediately close to shore, density is low; about 5 km from shore, density peaks; and by 30 km from shore, density tapers back to low (Figure 1).

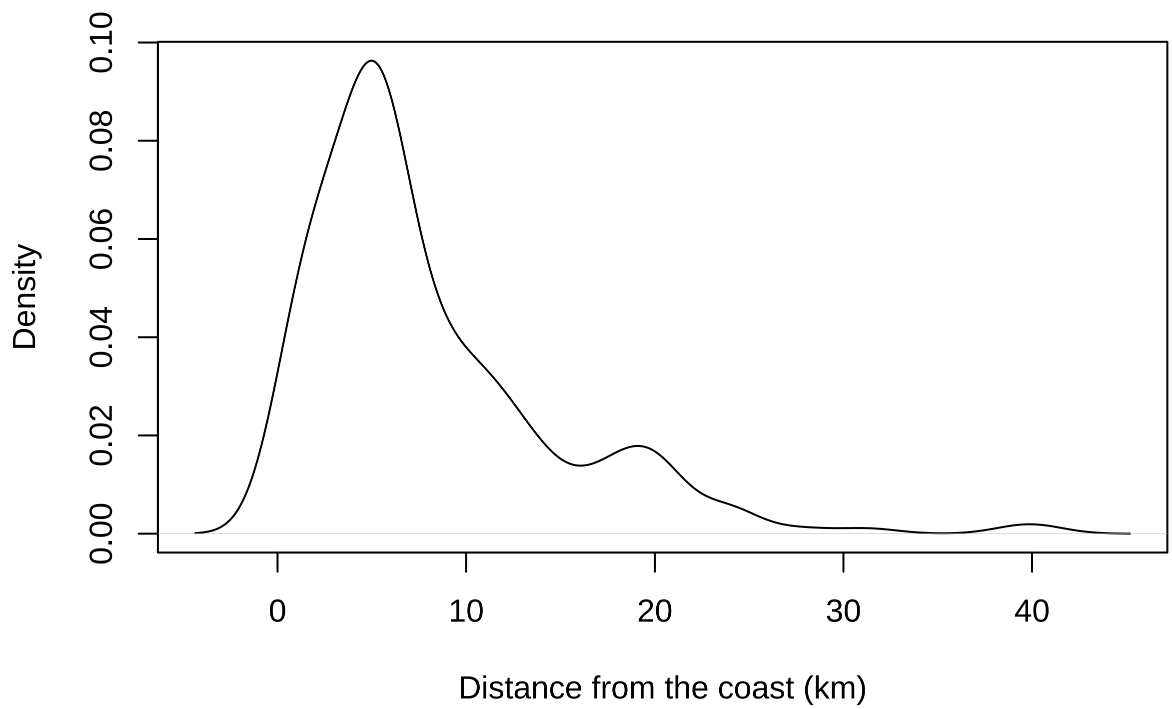

Figure 1: Density of Marbled Murrelet nests relative to straight-line distance from the coast (km).

Using the `{fitdistrplus}` package, use a Cullen & Frey diagram to determine what distribution best describes nest distance-from-the-coast distribution (Figure 2).

```
library(fitdistrplus)

# Find the best distribution
descdist(nests$nest_dist_km, discrete = FALSE, boot = 1000)
```

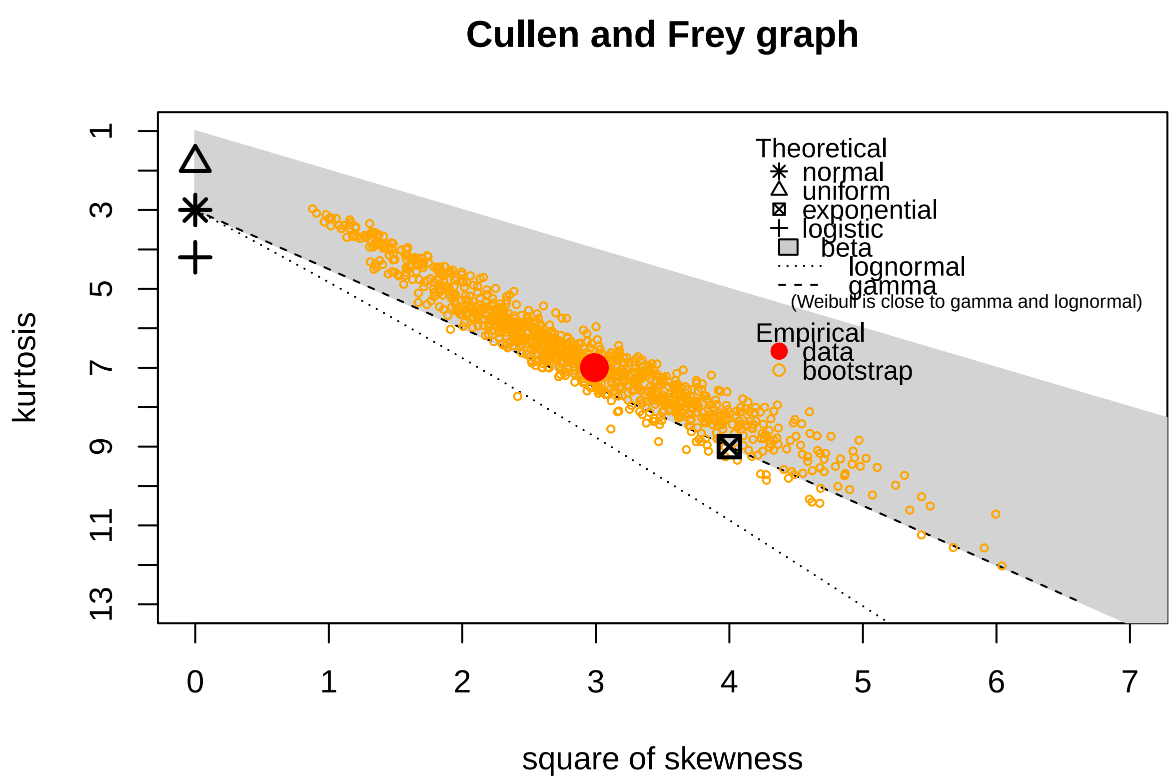

Figure 2: Skewness-kurtosis plot of nest distances from the shore in km.

```
## summary statistics
## ------
## min:  0.3535534   max:  40.4753
## median:  5.648556
## mean:  8.019207
## estimated sd:  6.67387
## estimated skewness:  1.728896
## estimated kurtosis:  6.996983
```

A gamma distribution best fits our count data (beta distributions are only suitable for data bounded from 0-1). Fit a gamma function to our empirical nest data and examine the fit (Figure 3).

```r
# Fit gamma distribution
fit <- fitdist(nests$nest_dist_km, distr = "gamma", method = "mle")

plot(fit)
```

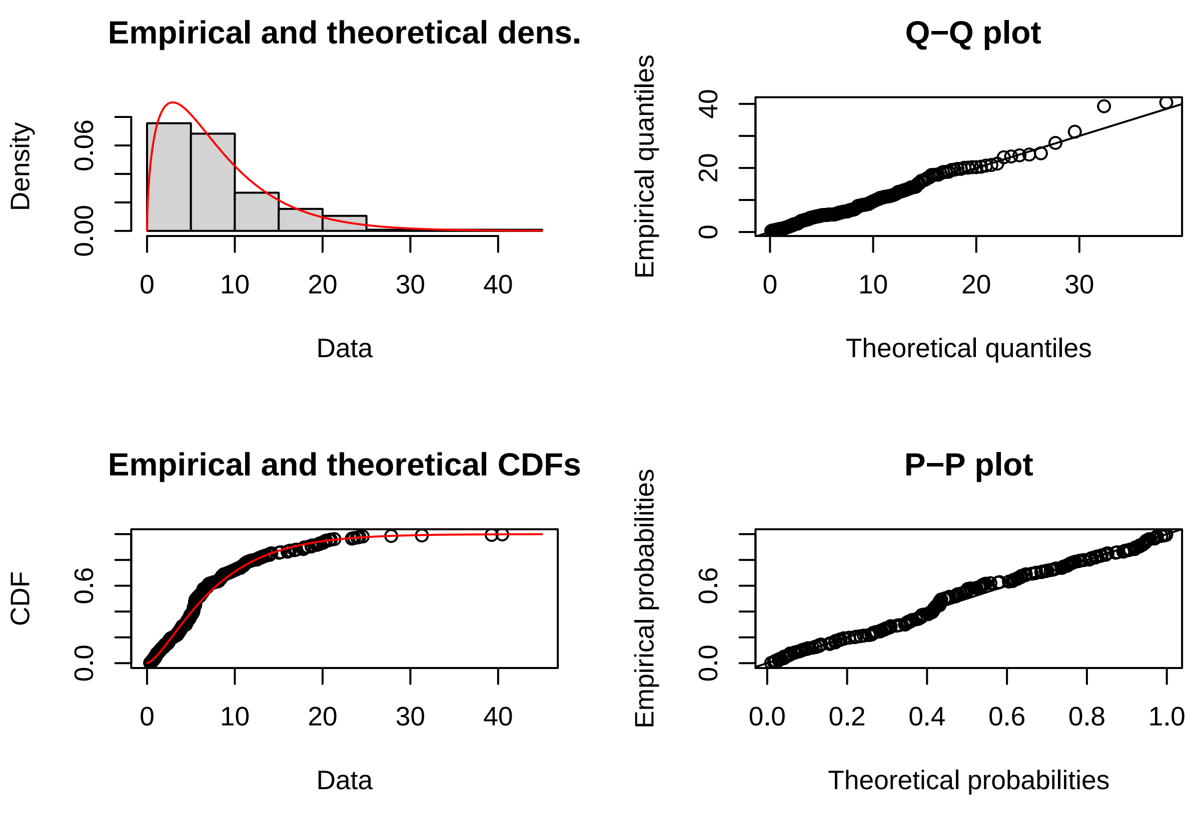

Figure 3: Nest gamma distribution fit diagnostics. The top-left plot shows the empirical distribution of nest data (grey bars) overlaid with the fitted density curve (red line). The x-axis is distance from the coast in km. The remaining plots show model fit diagnostics.

2. Spatialize the gamma distribution

Rasterize and normalize this curve, such that every pixel of the BC coastline runs from 0 (low/zero probability) to ~1 (high probability) of finding a Marbled Murrelet nest, based on the pixel's distance from the coast.

```r
library(terra)

ifel(sea_dist > 0, sea_dist, NA) |>
  plot(main = "Distance from coast (km)") # example coast distance raster
```

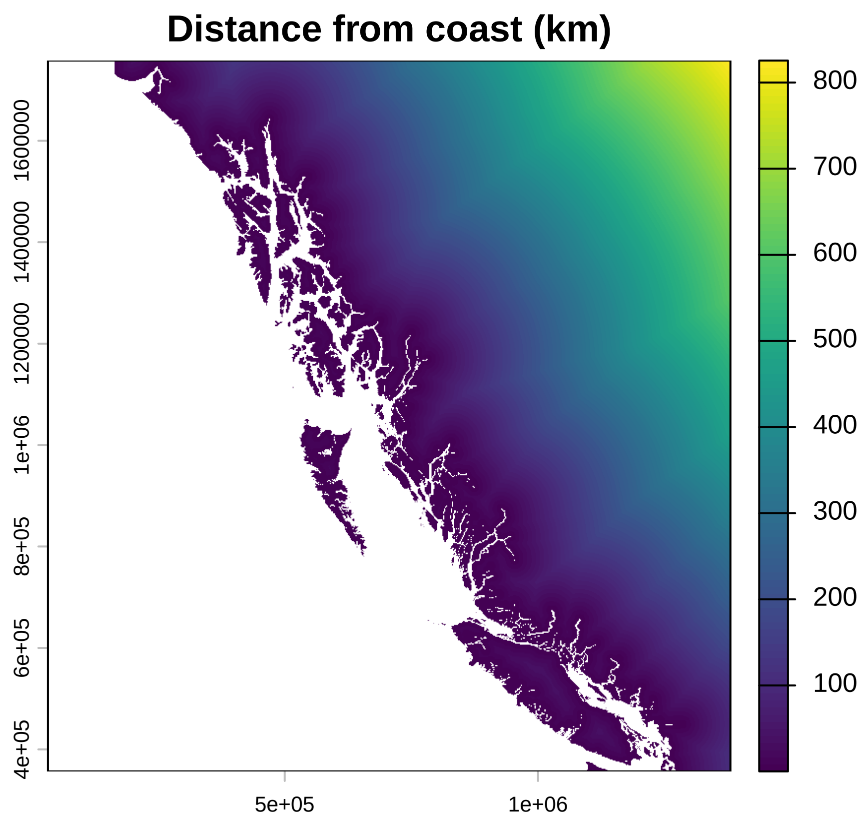

Figure 4: Map with pixels colored by the distance from the coast in km.

```r
# Replace distance from coast with gamma probabilities
# Derive a table of every distance with the gamma density
# function value at each distance
distances <- terra::values(sea_dist)
# Predict values at each distance following the fit gamma distr
g <- dgamma(distances,
            shape = fit$estimate[[1]],
            scale =  1 / fit$estimate[[2]])
# Normalize to 0-1
p <- g / max(g, na.rm = TRUE)
```

```r
# Now put the nest probability values into a raster
p <- terra::rast(vals = p,
                 crs = terra::crs(sea_dist),
                 terra::ext(sea_dist),
                 res = terra::res(sea_dist))

# Cut out ocean bits
p <- ifel(p == 0, NA, p)

# Plot
plot(p, main = "Nest probability")
```

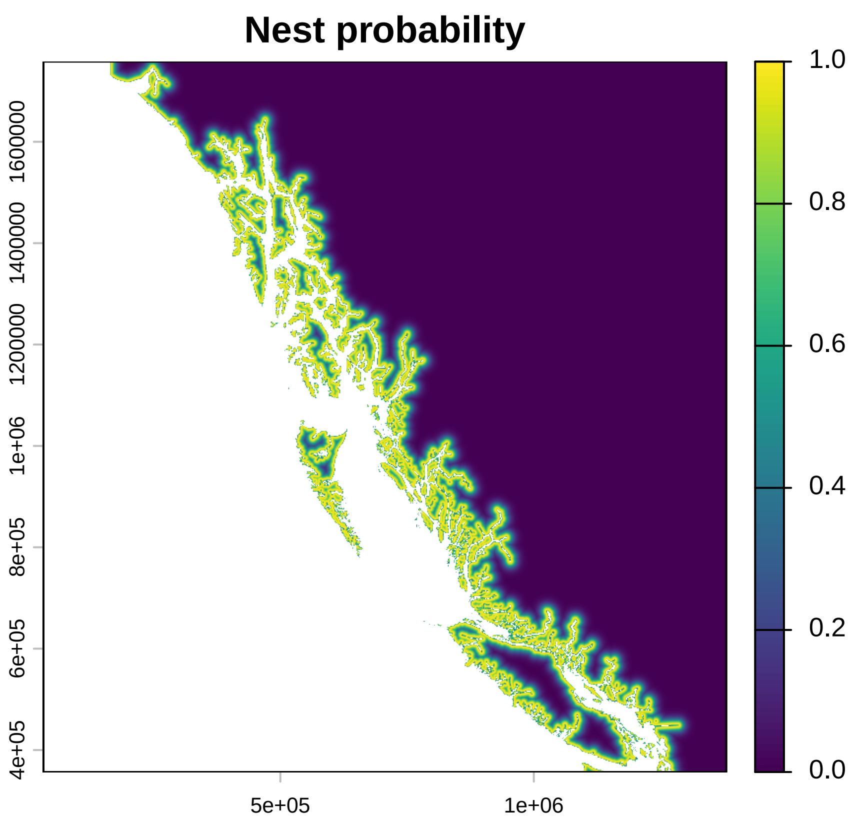

Figure 5: Rasterized gamma curve. Pixels are colored on a scale from 0 to 1, with higher numbers indicating a higher likelihood of nest occurrence. The value of the pixel is determined by the nest gamma distribution (Figure 2), with pixels closer to shore having a higher likelihood of nest occurrence. (Note that pixels immediately next to shore have a lower probability, with the peak in occurrence at around ~3 km from shore.)

Clip that nest probability raster to only occur within the suitable habitat layer.

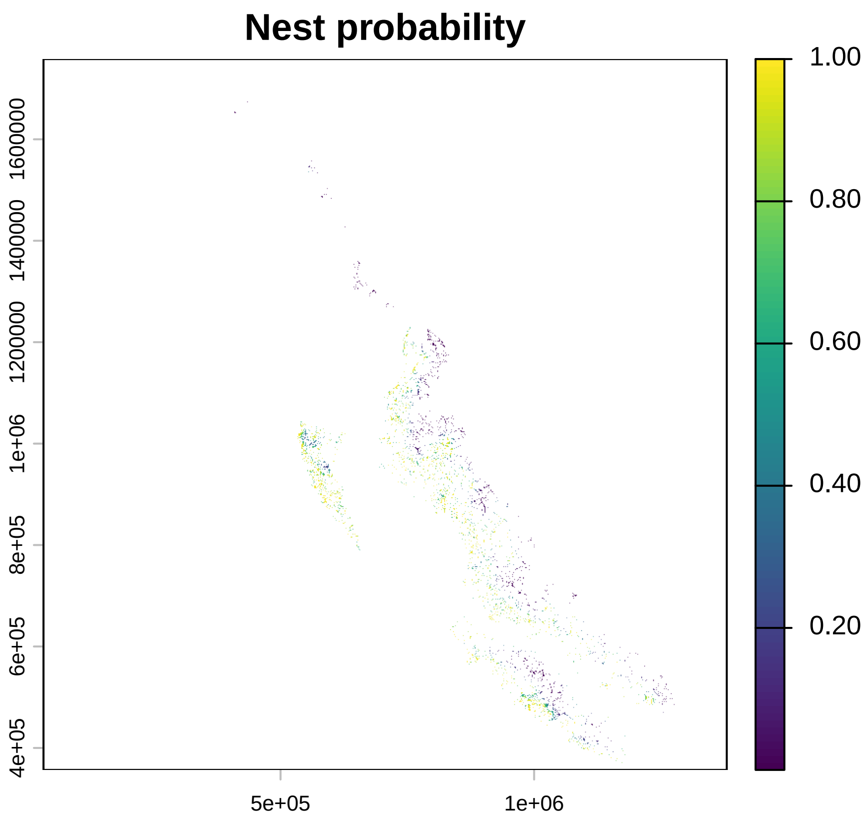

Figure 6: Rasterized gamma curve (Figure 4), clipped to Marbled Murrelet suitable habitat boundaries. This raster represents likelihood of nest occurrence.